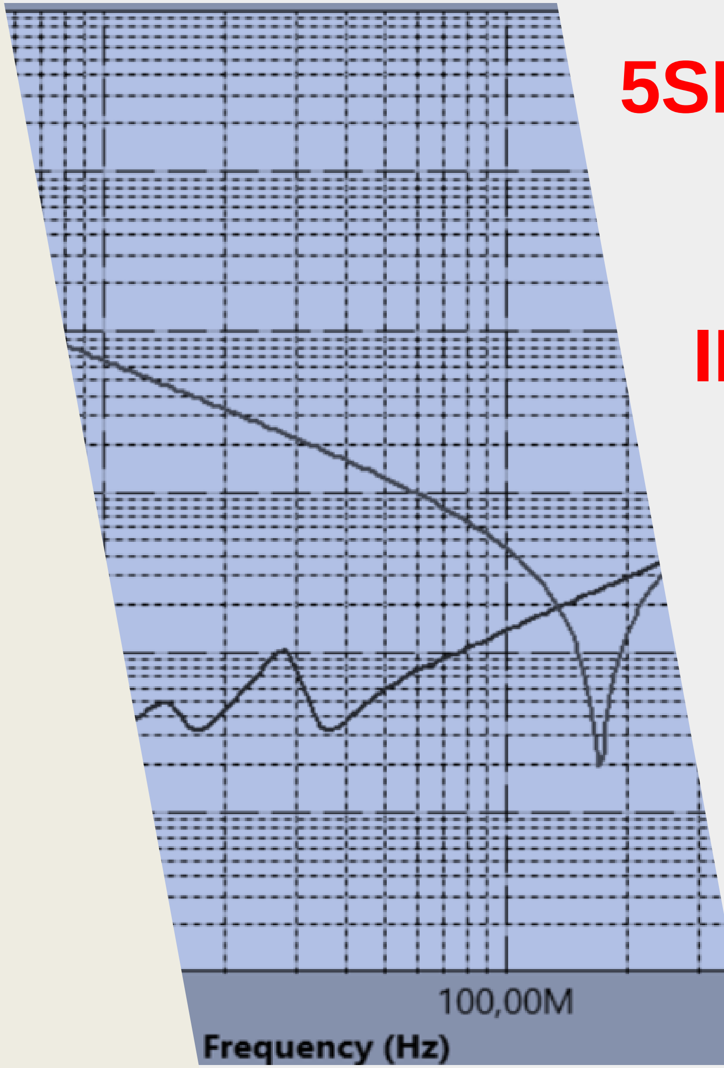

5SEE – MACHINE LEARNING FOR ELECTRONIC DESIGN

INTRODUCTION TO THE LAB: POWER INTEGRITY ISSUES

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November 2025

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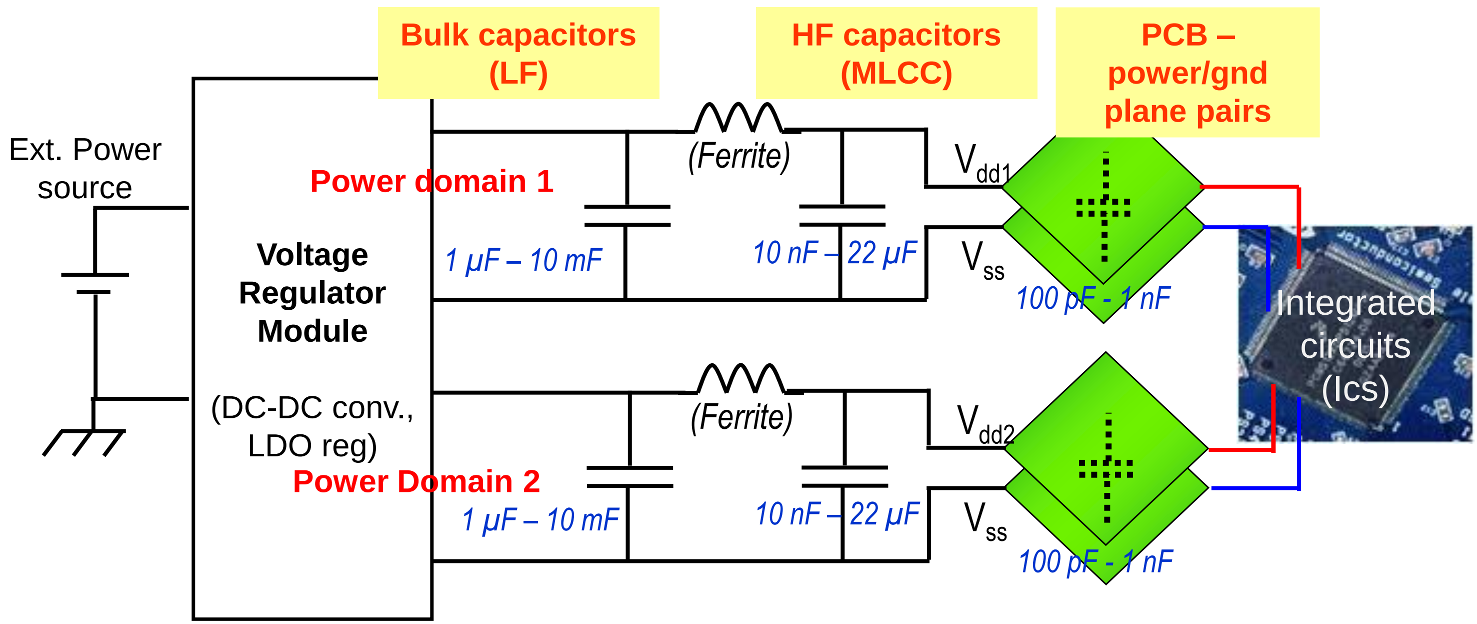
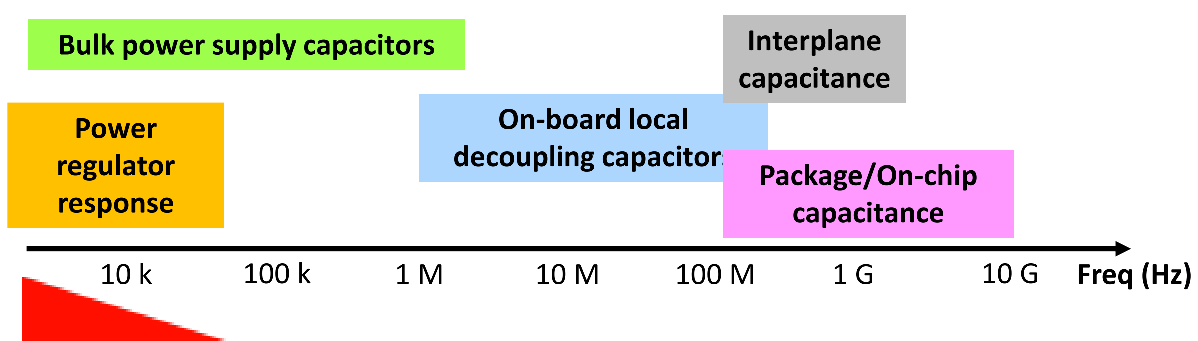
Power Distribution Network of PCB

- **Typical structure of multilayer PCB Power Distribution Network (PDN)**
  - ✓ Purpose: deliver stable power and ground references of digital ICs mounted on a PCB.



- ✓ Typical frequency range of influence of the typical elements of a multilayer PCB PDN :



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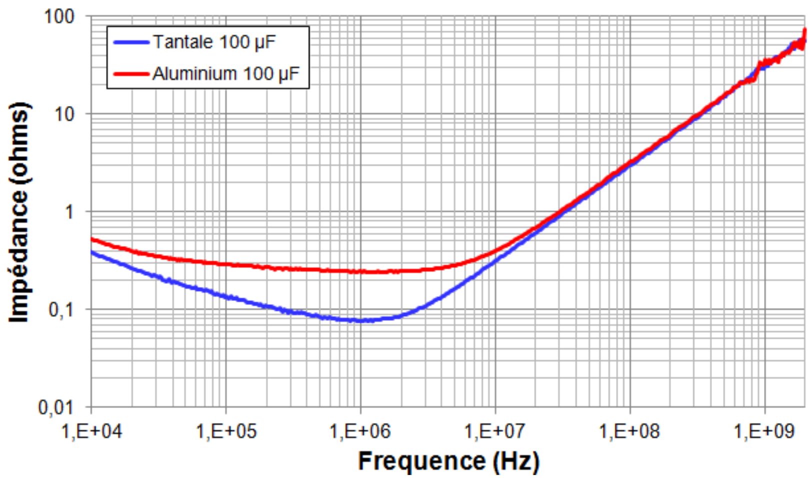
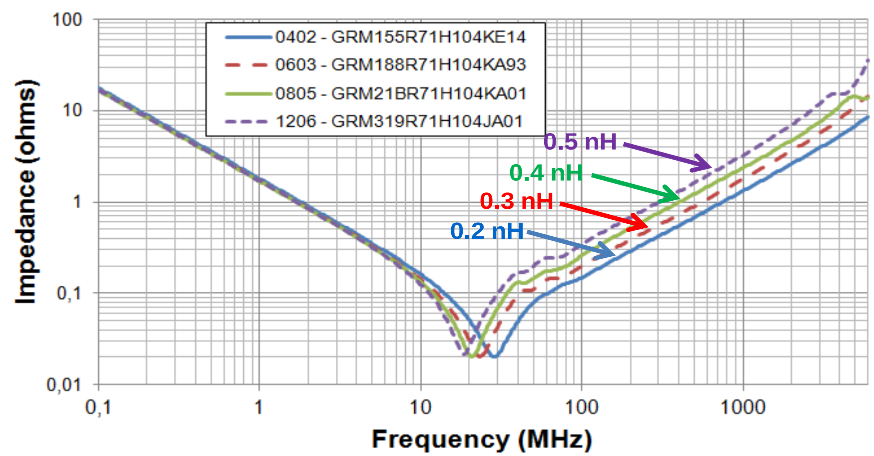
Power Distribution Network of PCB

Role of decoupling capacitors

- ✓ Local charge tank to respond rapidly to current peak produced by ICs.
- ✓ According to the targeted frequency range, different technologies/packages are used.
- ✓ The complex impedance Z(f) is the most important indicator → Z(f) small = efficient filtering !



Typical Z(f) for bulk capacitors (100 µF)

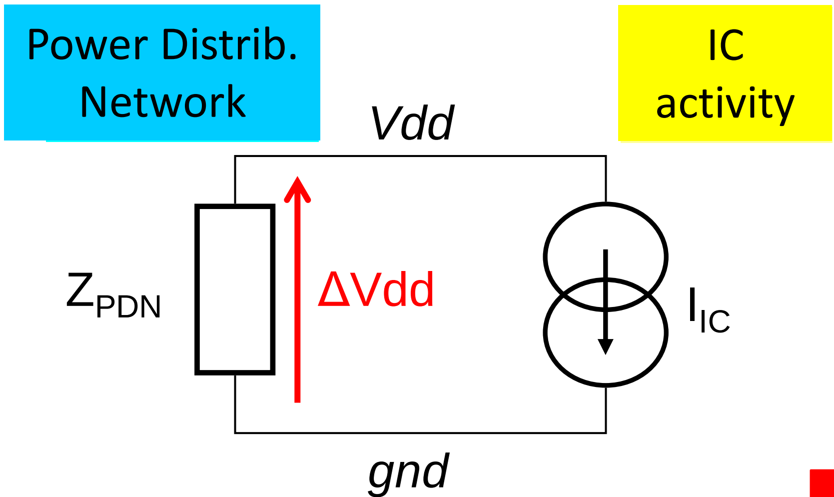


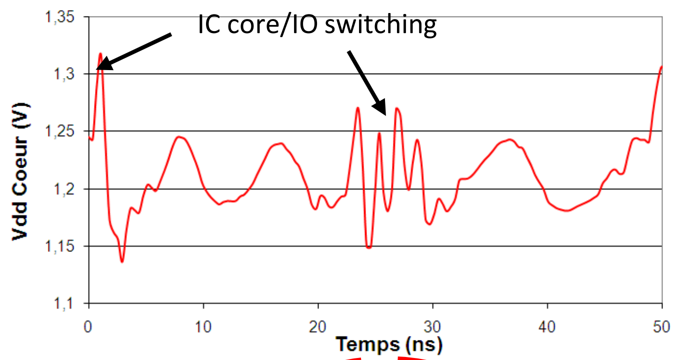
Typical Z(f) for MLCC (100 nF) mounted in different SMD cases

Power Integrity

Power integrity (PI) issues

- ✓ Small signal HF equivalent circuit of a digital IC mounted on a multilayer PCB – simulation of power supply voltage fluctuations due to IC transient current.





$$\Delta V_{dd}(f) = Z_{PDN}(f) \times I_{IC}(f)$$

Ensuring power integrity = keep power supply/ground voltage ΔVdd stable whatever the IC activity

To ensure PI, the PCB designer must guarantee a sufficienty low PDN impedance over the frequency range covered by IC transient activity: VRM module (LF), Bulk capacitors (LF), MLCC (MF, HF) Power/gound planes design (HF).

Power Integrity

Concept of target impedance Z_{Target}

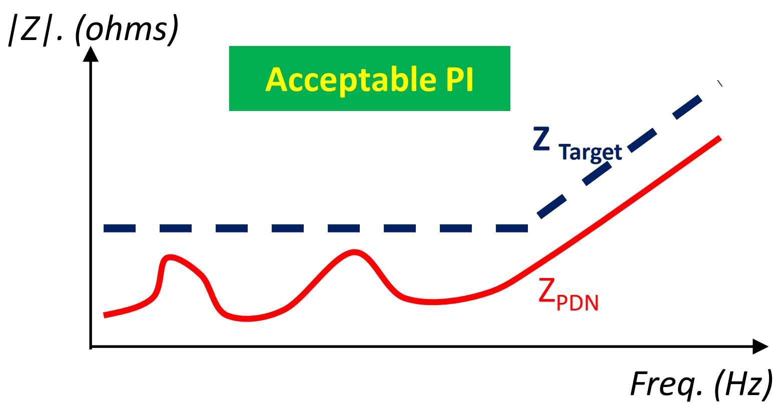
✓ Efficient figure of merit to ensure if a PDN design will ensure a sufficient PI.

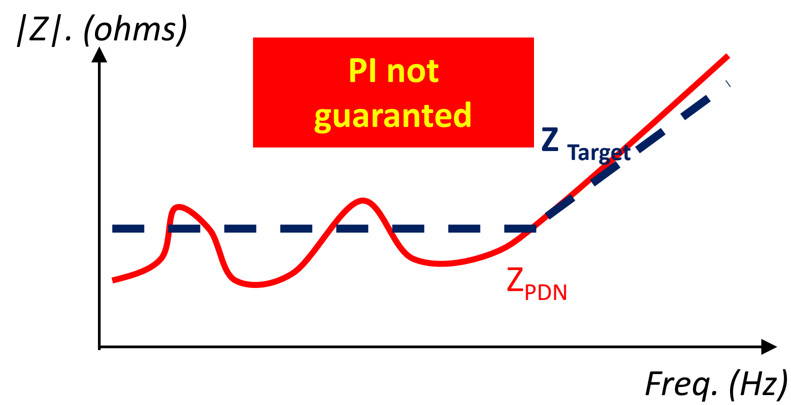
$$Z_{Target} = \frac{\Delta Vdd_{max}}{I_{avg}}$$

Max. allowed power supply voltage fluctuation (e.g. 5/10 % of Vdd)

Average current consumption (may be frequency-dependent)

✓ Design target:





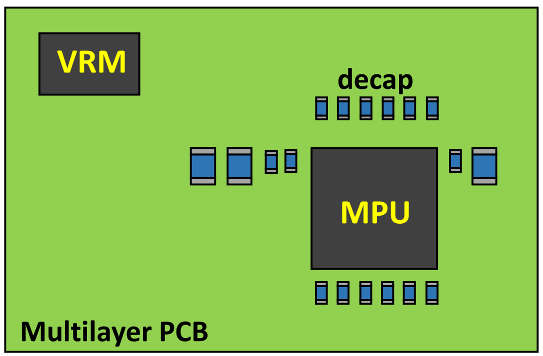
$$Z_{PDN}(f) < Z_{Target}(f) \ \forall f \in [f_{min} ; f_{max}]$$

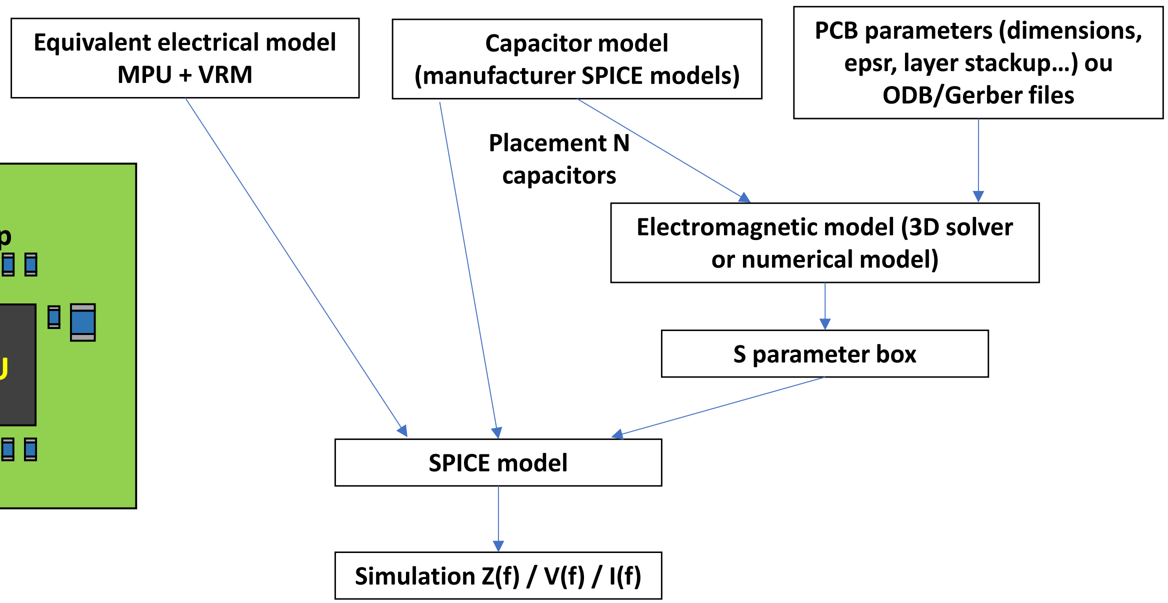
Select capacitors and placement + power/gnd plane pairs to ensure $Z_{PDN} < Z_{target}$.

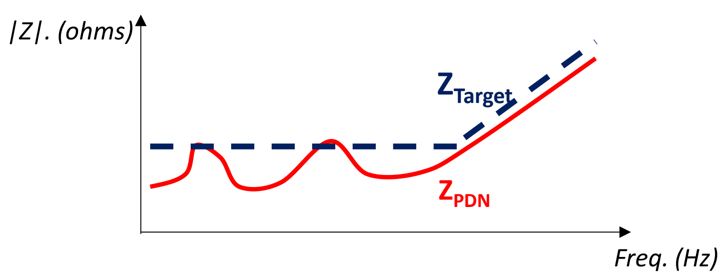
Power Integrity

Typical PI simulation flow

✓ For complex multilayer PCB with huge digital Ics (large transient signal), PI must be predicted to validate PDN design before fabrication.







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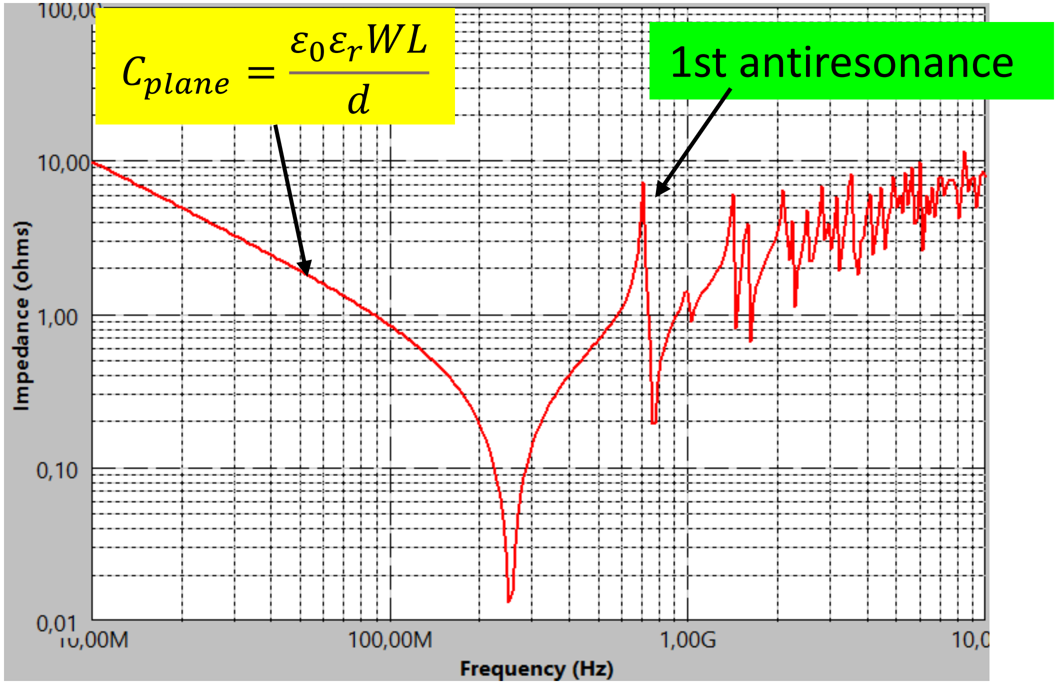
Power-ground plane response

Rectangular power-ground plane pair model

- ✓ A W×L rectangular power-ground plane pair separated by a short distance d forms a rectangular resonant cavity.
- ✓ If electrically short, equivalent to a plane capacitor.
- ✓ If electrically long, equivalent to a 2D transmission line terminated by open ends
- ✓ Response characterized by numerous resonant modes.

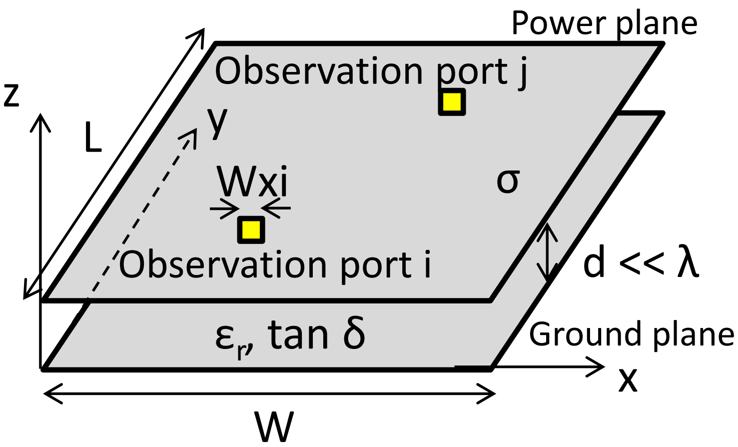


Simulated Z(f) of a power-ground plane pair (W=L=100 mm, d = 0.25 mm, εr = 4.5)

Power-ground plane response

Rectangular cavity model

- ✓ For rectangular plane pairs without any aperture, Z(f) can be determined analytically from Maxwell's equation resolution:



$$Z_{PGij}(\omega) = \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \frac{N'_{mni} N'_{mni}}{\frac{1}{j\omega L_{mn}} + j\omega C_{mn} + G_{mn}}$$

With:

$$N_{mni} = cos\left(\frac{m\pi x_i}{W}\right) cos\left(\frac{n\pi y_i}{L}\right) sin\,c\left(\frac{m\pi W_{xi}}{2W}\right) sin\,c\left(\frac{n\pi W_{yi}}{2L}\right)$$

$$L_{mn} = \frac{d}{2\pi F_{Cmn} W L \varepsilon_0 \varepsilon_r} \quad C_{mn} = \frac{W L \varepsilon_0 \varepsilon_r}{d} \quad N'_{mni} = C_m C_n N_{mni}$$

$$G_{mn} = \frac{W L \varepsilon_0 \varepsilon_r}{d} 2\pi F_{Cmn} \left(tan\,\delta + \frac{1}{d} \frac{1}{\sqrt{\pi F_{Cmn} \mu \sigma}}\right)$$

$$C_m, C_n = \begin{cases} 1 & if\ m, n = 0 \\ \sqrt{2} & otherwise \end{cases}$$

Can be estimated numerically by summing a finite number of resonant modes.

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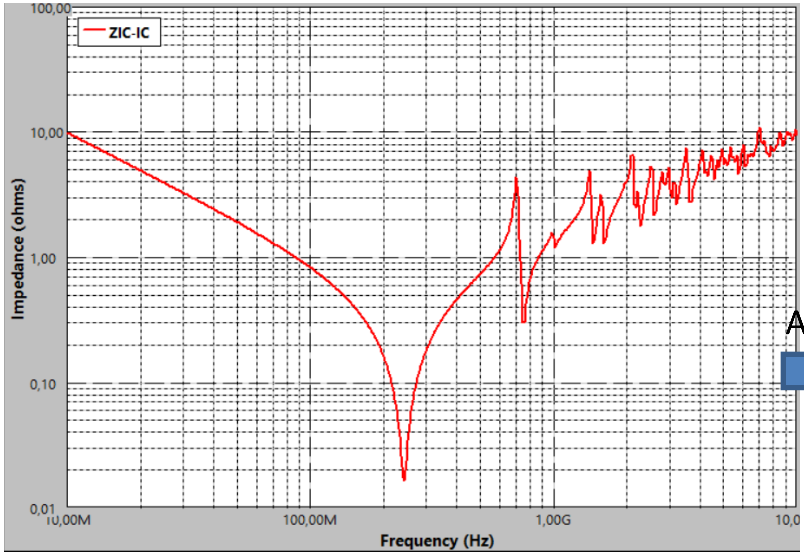
Power-ground plane response

Rectangular cavity model

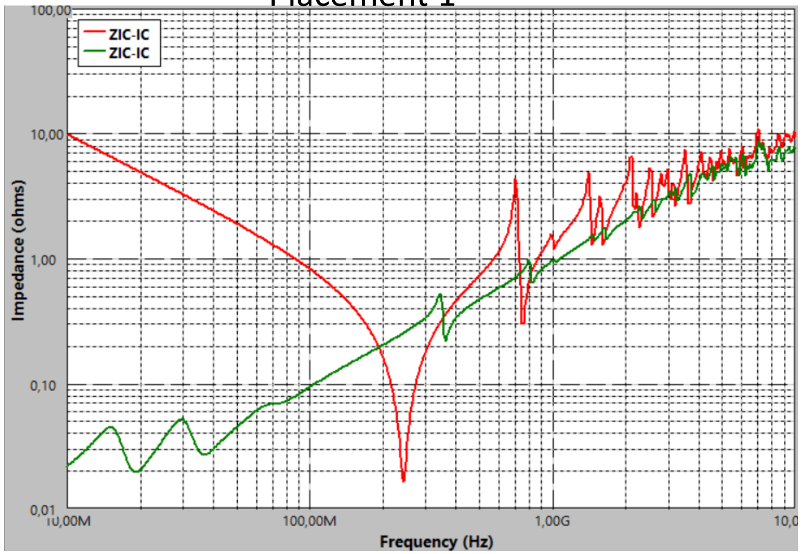
✓ Example: W=L=100 mm, d = 0.25 mm, εr = 4.5

Plane pair + IC + VRM (no decap)



Add 8 capacitors

Placement 1



Placement 2